CS610 SHORT NOTES FINAL TERM

LEC 23 TO 45

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Lecture No. 23

INTERNETWORKING: CONCEPTS, ARCHITECTURE AND PROTOCOLS

The Motivation for Internetworking

No single networking technology is best for all needs. A large organization with diverse networking requirements needs multiple physical networks. If the organization chooses the type of network that is best for each task, the organization will need several types of networks.

The interconnection of two or more networks, usually local area networks so that data can pass between hosts on the different networks as though they were one network. This requires some kind of Router or Gateway.

THE CONCEPT OF UNIVERSAL SERVICE:

The chief problems with multiple networks are as follows:

- A computer attached to a given network can only communicate with other computers attached to the same network.
- In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

UNIVERSAL SERVICES:

A communication system that supplies universal services allows arbitrary pairs of computers to communicate. Universal service is desirable because it increases individual productivity.

UNIVERSAL SERVICES IN A HETEROGENEOUS WORLD:

Extension techniques such as bridging cannot be used with heterogeneous network because of incompatible packet formats.

INTERNETWORKING:

Despite the incompatibilities among networks, researchers have devised a scheme

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that provides universal service among heterogeneous networks called 'internetworking'. It uses both hardware and software.

Physical Network Connection with Routers

A router is a special-purpose system dedicated to the task of interconnecting networks. A router can interconnect networks that use different technologies, including different media, physical addressing schemes, or frame formats.

Internet Architecture

Organizations seldom use a single router to connect its entire network for two reasons.

- Because the router must forward each packet, the processor in a given router is insufficient to handle the traffic.
- Redundancy improved internet reliability.

An Internet consists of a set of networks interconnected by routers. The internet scheme allows each organization to choose the number and type of network, the number of routers to use to interconnect them, and the exact interconnection topology.

Achieving Universal Services

The goal of internetworking is universal service across heterogeneous networks. To provide this service all computers and routers must agree to forward information from a source on one network to a specified destination. The task is complicated as frame formats and addressing schemes may differ. The key is universal protocol software (TCIP/IP).

A Virtual Network

A virtual network is shown in the figure that TCP/IP software provides to users and applications.

LAYERING AND TCP/IP PROTOCOLS: LONG

TCP/IP protocols are organized into five conceptual layers. Although some layers of the TCP/IP reference model correspond to layers of the ISO reference model,

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the ISO layers scheme does not have a layer that corresponds to TCP/IP Internet Layer.

Application Layer 5

Corresponds to basic network hardware layer in OSI..

Transport Layer 4

Specifies how to organize data in frames.

Internet Layer 3

Specifies The Format Of Packets Sent Across An Internet And Forward Packets.

Network Interface Layer 2

Specifies How To Ensure Reliable Transfer.

Physical Layer 1

Like 6 And 7 In OSI Species How One Application Uses An Internet.

Lecture No. 24 IP: INTERNET PROTOCOL ADDRESSES

Address for the Virtual Internet

To provide uniform addressing in internet protocol software defines an abstract addressing scheme that assigns each host a unique protocol address. Users, application programs, and higher layers of protocol software use the abstract protocol addresses to communicate.

The IP **Addressing** Scheme

An Internet Address (IP Address) is a unique 32-bit binary number assigned to a host and used for all communication with the host. Each packet sent across an internet contains the 32-bit IP address of the sender (source) as well as the intended recipient (destination).

The IP Address Hierarchy

Each 32-bit IP address is divided into two parts:

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- Prefix:
  identifies the physical network to which the computers are attached.
- Suffix:
  identifies an individual computer on the network.

The physical network in an internet is assigned a unique value known as a network number. No two networks can be assigned the same network number and no two computers on the same network can be assigned the same suffix. A suffix value can be used on more than one network.

The IP address hierarchy guarantees two important principles:

- Each computer is assigned a unique address.
- Although network number assignment must be coordinated globally, suffices can be assigned locally.

Original Classes of IP Addresses

The original IP address scheme divides host addresses into three primary classes. The class of an address determines the boundary between the network prefix and suffix.

Lecture No. 25 INTERNET PROTOCOL ADDRESS NOTATIONS

COMPUTING THE CLASS OF AN ADDRESS: LONG

Whenever it handles a packet, IP software needs to separate the destination address into a prefix and suffix. Classful IP addresses are self-identifying because the class of the address can be computed from the address itself.

First Four Bits Of Address	Table Index (in decimal)	Class of Address
0000	0	A
0001	1	A
0010	2	A
0011	3	A
0100	4	A
0101	5	A
0110	6	A
0111	7	A
1000	8	B
1001	9	B
1010	10	B
1011	11	B
1100	12	C
1101	13	C
1110	14	D
1111	15	E

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Dotted Decimal Notation

Dotted decimal notation is a syntactic form, the IP software uses to express 32-bit binary values when interacting with humans. Dotted decimal represent each octet in decimal and uses a dot to separate octets.

32-bit Binary Number	Equivalent Dotted Decimal
10000001 00110100 00000110 00000000	129.52.6.0
11000000 00000101 00110000 00000011	192.5.48.3
00001010 00000010 00000000 00100101	10.2.0.37
10000000 00001010 00000010 00000011	128.10.2.3
10000000 10000000 11111111 00000000	128.128.255.0

CLASSES AND DOTTED DECIMAL NOTATION:

The relationship between classes and dotted decimal notation is given as follows.

- In class A the three octets correspond to a host suffix.
- In class B the last two octets are the host octets.
- Class C has only one octet to represent the host.

Division of the Address Space

The IP class scheme does not divide the 32-bit address space into equal size class and the classes do not contain the same number of networks. A prefix of n bits allows unique network numbers, while a suffix of n bits allows host numbers to be assigned on a given network.

Authority for Addresses

Throughout the internet, each network prefix is unique. Networks obtain their network numbers from their Internet Service Provider (ISP). The Internet service providers coordinate with The Internet Assigned Number Authority to obtain their network numbers.

What is classful and classless addressing?

Classful addressing is a technique of allocating IP addresses that divides them into five categories. Classless addressing is a technique of allocating IP addresses that is intended to replace classful addressing in order to reduce IP address depletion.

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Lecture No. 26 IP SUBNETING

SUBNETS AND CLASSLESS ADDRESSING: LONG

As the Internet grew, the original Classful addressing scheme became a limitation. The IP address space was being exhausted because all networks had to choose one of three possible sizes. Many addresses were unused. Two new mechanisms were invented to overcome the limitations, which are as follows:

- Subnet addressing
- Classless addressing

Instead of having three distinct address classes, allow the division between prefix and suffix to occur on an arbitrary boundary. The classless addressing scheme solves the problem by allowing an ISP to assign a prefix that is, 28 bits long (allowing the host to have up to 14 hosts).

CIDR NOTATION:

Inside a computer, each address mask is stored as a 32-bit value. When we enter a prefix and an address mask they use a modified form of dotted decimal addressing called CIDR addressing, which is known as CIDR Notation.

CIDR HOST ADDRESSES:

CIDR IP addresses consist of two groups of numbers, which are also referred to as groups of bits. The most important of these groups is the network address, and it is used to identify a network or a sub-network (subnet). The lesser of the bit groups is the host identifier.

Lecture No. 27 ADDRESS REOLUTION PROTOCOL (ARP)

PROTOCOL ADDRESSES AND PACKET DELIVERY:

Protocol addresses are abstractions provided by software. Physical network hardware does not know how to locate a computer from it protocol address. The protocol address of the next hop must be translated to an equivalent hardware address before a packet can be sent.

Address Resolution LONG

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Mapping between a protocol address and a hardware address is called Address Resolution. A host or router uses address resolution when it needs to send a packet to another computer on the same physical network. A computer never resolves the address of a computer that attaches to a remote network.

Address resolution algorithms can be grouped into three basic categories:

- Table lookup
- Closed-form computation
- Message Exchange

Table lookup

Binding or mappings are stored in a table in memory, which the software searches when it needs to resolve an address.

Closed-form computation

The protocol address assigned to a computer is chosen carefully so that computer's hardware can be computed from the protocol address using basic Boolean and arithmetic operations.

Message Exchange

Computers exchange messages across a network to resolve an address. One computer sends a message that requests an address binding (translation) and another computer sends a reply that contains the requested information.

Address Resolution with Table Lookup LONG

Resolution requires a data structure that contains information about address binding. A separate address binding table is used for each physical network. The chief advantage of the table look up approach is generality; a table can store the address bindings for an arbitrary set of computers. For less than a dozen hosts, a sequential search can suffice. For large networks the sequential approach uses too much CPU time.

Two standard implementations to improve computational efficiency:

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- Hashing
- Direct Indexing

Hashing

Hashing is the transformation of a string of characters into a usually shorter fixed-length value or a key that represents the original string. Hashing is used to index and retrieve items in a database because it is faster to find the item using the shorter hashed key than to find it using the original value. It is also used in many encryption algorithms.

Direct Indexing

Less generally known technique. Possible only in cases where protocols addresses are assigned from a compact range.

Address Resolution with Closed-Form Computation LONG

For networks use configurable addressing it is possible to choose an address that make closed-form address resolution possible. A resolver is used to compute a mathematical function that maps an IP address to a hardware address. When a computer connects to a network that uses configurable address, the local network administrator must choose a hardware address as well as an IP address. The two values can be chosen to make address resolution trivial.

Address Resolution with Message Exchange LONG

An alternative to local computations is a distributed approach. A computer that needs to resolve an address sends a message across a network and receives a reply. The message carries a request that species the protocol address, and reply carries the corresponding hardware address.

Two possible designs:

- Centralized
- Distributed

CENTRALIZED

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A network includes one or more servers that are assigned the task of answering address resolution requests.

Advantage:

Resolution easier to configure, manage and control.

DISTRIBUTED

Each computer on the network participates in address resolution by agreeing to answer resolution request for its address.

Advantage:

Address resolution servers can become a bottleneck and reduce cost.

Lecture No. 28 ARP MESSAGE FORMAT

Address Resolution Protocol (ARP)

TCP/IP can use any of the three address resolution method depending on the addressing scheme used by the underlying hardware. To guarantee that all computers agree on the exact format and meaning of message used to resolve addresses, the TCP/IP protocol suite includes an Address Resolution Protocol (ARP).

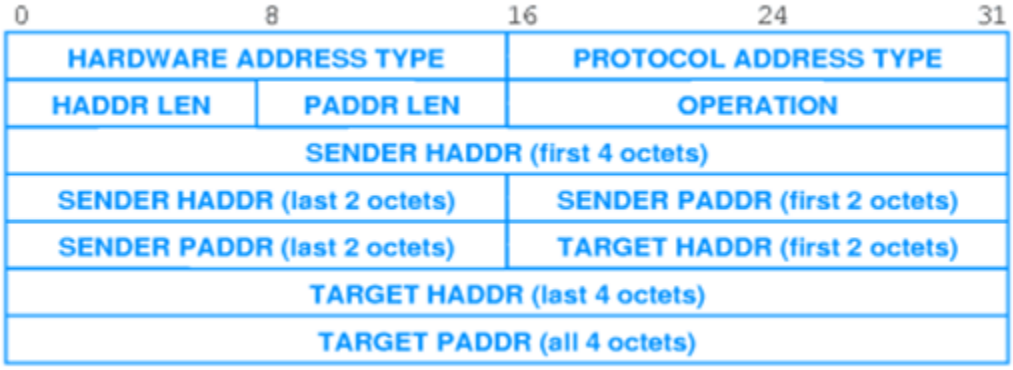
The ARP standard defines two basic message types:

- Request
  Contains an IP address and requests the corresponding hardware address.
- Response
  Contains both the IP address sent in the request and the hardware address.

ARP Message Format

Although the ARP message format is sufficiently general to allow arbitrary protocol and hardware addresses. ARP almost always used to bind a 32-bit IP address to a 48-bit Ethernet address.

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Sending an ARP Message

Encapsulation

When one computer sends an ARP message to another, the message travels inside the hardware frame. Technically, placing a message inside a frame for transport is called encapsulation.



CASHING ARP RESPONSES:

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient. To reduce network traffic, ARP software extracts and saves the information from a response so it can be used for subsequent packets. ARP manages the Table as cache short term storage.

PROCESSING AN INCOMING ARP MESSAGE:

When an ARP message arrives, the protocol specifies that the receiver must perform two basic steps:

- First, the receiver extracts the sender's address binding and checks to see if it is present in the cache. If not, it updates the cache.

- The receiver examines the operation field of the message to determine whether the message is a request or a response.
    - If the message is a request, the receiver compares the field TARGET PADDR with the local protocol address.
    - If the two are identical, the computer is the target of the request and must send an ARP response.

LAYERING, ADDRESS RESOLUTION AND PROTOCOL ADDRESSES:

Address resolution (ARP) is a network interface layer function. Protocol addresses used in all higher layers. Hides ugly details and allows generality in upper layers.

Lecture No. 29 IP DATAGRAMS AND DATAGRAM FORWARDING

Connectionless service

End-to-end delivery service is connectionless. Extension of LAN abstraction, Universal addressing, Data delivered in packets (frames), each with a header. Combines collection of physical networks into single, virtual network.

Transport protocols use this connectionless service to provide:

- Connectionless data delivery (UDP)
- Connection-oriented data delivery (TCP)

VIRTUAL PACKETS:

These packets serve same purpose in Internet as frames on LAN. Each packet has a header. Routers, which are formally gateways, forward packets between physical networks. Virtual packets include packets at any layer or sub layer (as those terms are used in, for example, the OSI model) above the most basic packets or frames used in a network. These "virtual packets" allow heterogeneous networks to talk to each other using a common protocol.

IP DATAGRAM FORMAT:

The unit of IP data delivery is called a Datagram. It includes header area and data area.

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IP DATAGRAM SIZE: LONG

Datagrams can have different sizes:

- Header area usually fixed (20 octets) but can have options.
- Data area can contain between 1 octet and 65,535 octets (216 - 1).
- Usually, data area is much larger than header.

Forwarding Datagrams LONG

Header contains all information needed to deliver datagram to the destination computer.

- Destination address
- Source address
- Identifier
- Other delivery information
- Router examines header of each datagram and forwards datagram along path to destination.

ROUTING TABLE:

A routing table contains the information necessary to forward a packet along the best path toward its destination. Each packet contains information about its origin and destination. Routing Table provides the device with instructions for sending the packet to the next hop on its route across the network.

ROUTING TABLES AND ADDRESS MASKS:

Additional information is kept in routing table. Destination is stored as network address. Next hop is stored as IP address of router. Address mask defines how many bits of address are in prefix. Prefix defines how much of address used to identify network.

Address Masks

To identify destination network, apply address mask to destination address and compare to network address in routing table. Can use Boolean

if ((Mask[i] & D) == Dest[i]) forward to NextHop[i]

ORWARDING, DESTINATION ADDRESS AND NEXT-HOP:

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Destination address in IP datagram is always ultimate destination. Router looks up next-hop address and forwards datagram.

Network interface layer takes two parameters:

- IP datagram
- Next-hop address

Next-hop address never appears in IP datagram.

BEST-EFFORT DELIVERY:

IP provides service equivalent to LAN. Does not guarantee to prevent:

- Duplicate datagrams
- Delayed or out-of-order delivery
- Corruption of data
- Datagram loss

Lecture No. 30 IP ENCAPSULATION, FRAGMENTATION AND REASSEMBLY

IP DATAGRAM OPTIONS:

Several options can be added to IP header Record route Source route Time stamp. Header with no options has H. LEN field value 5; data begins immediately after DESTINATION IP ADDRESS. Options added between DESTINATION IP ADDRESS and data in multiples of 32 bits. Header with 96 bits of options has H. LEN field value 8.

Encapsulation

Network interface layer encapsulates IP datagram as data area in hardware frame. Hardware ignores IP datagram format and Standards for encapsulation describe details. Standard defines data type for IP datagram, as well as others (e.g., ARP). Receiving protocol stack interprets data area based on frame type.

ENCAPSULATION ACROSS MULTIPLE HOPS:

Each router in the path from the source to the destination un-encapsulates incoming datagram from frame, processes datagram and determines next hop and

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encapsulates datagram in outgoing frame. Datagram may be encapsulated in different hardware format at each hop. Datagram itself is (almost) unchanged.

Maximum Transmission Unit MTU: LONG

Every hardware technology specification includes the definition of the maximum size of the frame data area called the Maximum Transmission Unit (MTU). Any datagram encapsulated in a hardware frame must be smaller than the MTU for that hardware.

MTU and Heterogeneous Networks

An Internet may have networks with different MTUs. Downstream network has smaller MTU than local network.



Fragmentation: LONG

One technique limits datagram size to smallest MTU of any network. IP uses fragmentation datagrams can be split into pieces to fit in network with small MTU. Router detects datagram larger than network MTU Splits into pieces and each piece smaller than outbound network MTU.

DATAGRAM REASSEMBLY:

Reconstruction of original datagram is called reassembly.

FRAGMENT LOSS:

Fragmentation is done by the network layer when the maximum size of datagram is greater than maximum size of data that can be held in a frame i.e., its Maximum Transmission Unit (MTU). The network layer divides the datagram received from the transport layer into fragments so that data flow is not disrupted.

Fragmenting a Fragment

Fragment may encounter subsequent network with even smaller MTU. Router fragments the fragment to fit. Resulting sub-fragments look just like original

fragments (except for size).No need to reassemble hierarchically; sub-fragments include position in original datagram.

Lecture No. 31 THE FUTURE IP (IPV6)

SUCCESS OF IP:

IP has accommodated dramatic changes since original design. But basic principles are still appropriate today. There are many new types of hardware.

SCALING:

Scale is also dramatically changed. Size from a few tens to a few tens of millions of computers has been revolutionized. Speed has increased from 56Kbps to 1Gbps. Also there is an increased frame size in hardware.

MOTIVATION FOR CHANGE: LONG

- One of the parameters, which motivated IP for change, is address space. The 32- bit address space allows for over a million networks. But most networks are class C and too small for many organizations. 214 class B network addresses already almost exhausted (and exhaustion was first predicted to occur, a couple of years ago).
- The second parameter is type of service, the IP provides. Different applications have different requirements for delivery reliability and speed. Current IP has type of service that is not often implemented. Another factor for the motivation for change is multicast.

NAME AND VERSION NUMBER: LONG

A preliminary version of IP was called IP- Next Generation (IPng). There were several proposals and all called IPng. One name was selected and it used next available version number i.e. 6. The result is IP version 6 (IPV6).

NEW FEATURES: LONG

The new features of IPV6 are as follows:

- IPV6 addresses are 128 bits.
- Header format is entirely different.
- Additional information is stored in optional extension headers, followed by data.
- Flow label and quality of service allows audio and video applications to establish appropriate connections.
- New features can be added more easily. So it is extensible.

PARSING IPv6 HEADERS: LONG

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Base header is fixed size i.e. 40 octets. NEXT HEADER field in the base header defines type of header and it appears at end of fixed-size base header. Some extension headers are variable sized. NEXT HEADER field in extension header defines type. HEADER LEN field gives size of extension header.

Lecture No. 32 IPv6 AND AN ERROR REPORTING MECHANISM

FRAGMENTATIO LONG .

Fragmentation information is kept in separate extension header. Each fragment has base header and (inserted) fragmentation header. Entire datagram including original header may be fragmented.

FRAGMENTATION AND PATH MTU:

IPv6 source (not intermediate routers) is responsible for fragmentation. Routers simply drop datagrams larger than network MTU (Maximum Transmission Unit). So source must fragment datagram to reach destination. Source determines path MTU. The smallest MTU on any network between source and destination and it fragments datagram to fit within that MTU.

The process of learning the path MTU is known as path MTU discovery. Path MTU discovery is used. Source sends probe message of various sizes until destination reached. It must be dynamic i.e. path may change during transmission of datagrams.

USE OF MULTIPLE HEADERS:

It has following advantages:

- Efficiency:
  Header is only as large as necessary.
- Flexibility:
  it can add new headers for new features.
- Incremental development:
  It can add processing for new features to testbed, other routers will skip those headers.

IPv6 ADDRESSING: LONG

IPv6 uses 128-bit addresses. A 128-bit address includes network prefix and host suffix. An advantage of IPv6 addressing is that it has no address classes i.e. prefix/suffix boundary can fall anywhere.

Following are special types of addresses, IPv6 uses:

- Unicast:
  It is used for single destination computer.
- Multicast:
  It is used for multiple destinations; possibly not at same site.
- Cluster:
  This type of address is used for collection of computers with same prefix, datagram is delivered to one out of cluster.

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IPv6 ADDRESS NOTATION: LONG

128-bit addresses unwidely in dotted decimal; requires 16 numbers:
105.220.136.100.255.255.255.255.0.0.18.128.140.10.255.255

Groups of 16-bit numbers in hex separated by colons – colon hexadecimal (or colon hex). 69DC: 8864:FFFF: FFFF: 0:1280:8C0A:FFFF

Zero-compression – series of zeroes indicated by two colons FF0C: 0:0:0:0:0:0:B1 FF0C::B1

IPv6 address with 96 leading zeros is interpreted to hold an IPv4 address.

INTERNET CONTROL MESSAGE PROTOCOL: LONG

Some errors can be reported. Router sends message back to source in datagram. Message contains information about problem. It is encapsulated in IP datagram.

TYPES OF MESSAGES:

Internet control Message Protocol (ICMP) defines error and informational messages. These are given as follows:

1. ERROR MESSAGES:
   These are as follows:
   - Source quench
   - Time exceeded
   - Destination unreachable
   - Redirect
   - Fragmentation required
2. INFORMATIONAL MESSAGES:
   These are as follows:
   - Echo request/reply
   - Address mask request /reply
   - Router discovery

Lecture No. 33 AN ERROR REPORTING MECHANISM (ICMP)

ICMP MESSAGE TRANSPORT:

ICMP message transport is acted upon by getting ICMP encapsulated in IP. ICMP message is sent in response to incoming datagrams with problems. ICMP message is not sent for ICMP message.

USING ICMP TO TEST REACHABILITY: LONG

ICMP can also be used to test different tools. An Internet host A, is reachable from another host B, if datagrams can be delivered from A to B. Ping program tests reachability. It sends datagram from B to A,

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which echoes back to B. it uses ICMP echo request and echo reply messages. Internet layer includes code to reply to incoming ICMP echo request messages.

USING ICMP TO TRACE A ROUTE: LONG

A trace route works by sending Internet Control Message Protocol (ICMP) packets, and every router involved in transferring the data gets these packets. The ICMP packets provide information about whether the routers used in the transmission are able to effectively transfer the data.

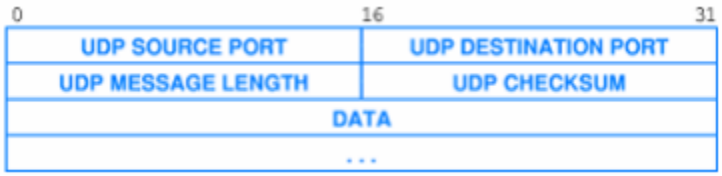
THE LAST ADDRESS PRINTED BY TRACE-ROUTE: LONG

There are two possibilities used to detect the destination.

- Send and ICMP echo request, destination host will generate an ICMP echo reply.
- Send a datagram to a non-existent application, destination host will generate an ICMP destination unreachable message.

Lecture No. 35 DATAGRAM FORMAT AND TCP: RELIABLE TRANSPORT SERVICE

UDP DATAGRAM FORMAT:



TCP: LONG

TCP is the major transport protocol in the TCP/IP suite. It uses unreliable datagram service offered by IP when sending data to another computer. It provides reliable data delivery service to applications.

THE NEED FOR RELIABLE TRANSPORT:

Reliability is fundamental in a computer system. Software in the Internet must provide the same level of reliability as a computer system. Software must guarantee prompt and reliable communication without any loss, duplication, and change in the order.

TRANSMISSION CONTROL PROTOCOL:

Reliability is the responsibility of the Transport layer. In TCP/IP, TCP provides reliable transport service. Most Internet applications use TCP as no other protocol has proved to work better.

SERVICE PROVIDED BY TCP:

Following are the services provided by TCP:

- Connection-oriented service
- Point-to-point
- Complete reliability
- Full-duplex communication
- Stream interface
- Reliable connection startup
- Graceful connection shutdown

END-TO-END SERVICE AND DATAGRAMS: long

Applications can request a connection. TCP connections are called Virtual Connections. They are created by software only. Internet does not provide software or hardware support for the connections. TCP software modules on two computers create an illusion of a connection. TCP uses IP to carry messages. TCP message is encapsulated in IP datagram and sent to the destination. On the destination host, IP passes the contents to TCP.

ACHIEVING RELIABILITY:

The major problems in the reliable delivery are:

- Unreliable delivery by the underlying communication system.
- System reboots.

Lecture No. 36 TCP: RELIABLE TRANSPORT SERVICE (Cont.)

PACKET LOSS AND RETRANSMISSION:

TCP achieves reliability by retransmission. An acknowledgement is used to verify that data has arrived successfully. If acknowledgement does not arrive, the previous data is retransmitted.

HOW LONG SHOULD TCP WAIT BEFORE RETRANSMITTING:

The time for acknowledgement to arrive depends on:

- Distance to destination
- Current traffic conditions

THREE-WAY HANDSHAKE TO BEGIN A CONNECTION:

Part of the 3-way handshake used to create a connection, requires each end to generate a random 32-bit sequence number. If an application attempts to establish a new TCP connection after a computer reboots, TCP chooses a new random number.

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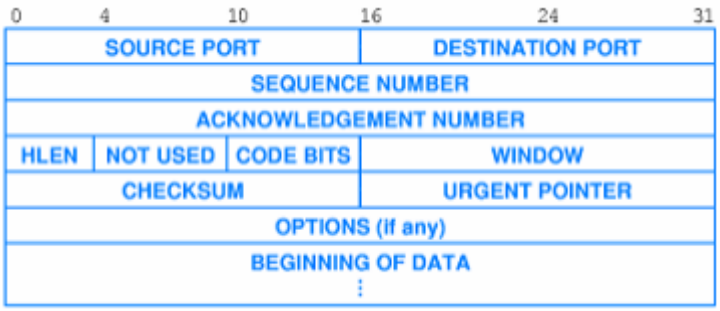
Lecture No. 37 NETWORK ADDRESS TRANSLATION (NAT)

CONGESTION CONTROL:

The goal of congestion control is to avoid adding retransmissions to an already congested network. Reducing the window size quickly in response to the lost messages does it. It is assumed that loss is due to congestion.

TCP SEGMENT FORMAT:

It is shown in the figure below. TCP uses single format for all messages. TCP uses the term segment to refer to a message. Each message sent from TCP on one machine to TCP on another machine uses this format including data and acknowledgement.



Lecture No. 38 NETWORK ADDRESS TRANSLATION

VARIANTS OF NAT: LONG

There are also some variants of NAT due to some of its drawbacks. The basic NAT simply changes IP addresses. But Network Address and Port Translation (NAPT) (which is another modified form of NAT) changes IP addresses and protocol port numbers too. It is the most popular form of NAT. Twice NAT is another variant of NAT. it is used with site that runs server. In this process NAT box is connected to Domain Name.

NETWORK ADDRESS AND PORT TRANSLATION (NAPT):

It is by far the most popular form of NAT that can change TCP or DP protocol port numbers as well as IP addresses. It allows multiple computers at site to communicate with single destination as well as multiple users on given computer to communicate with same destination.

TCP SPLICING:

A popular use of NAPT is TCP Splicing. It interconnects two independent TCP connections and performs segment rewriting. It is extremely efficient and avoids overhead of extracting data from one connection and sending to the other. It uses extended translation table.

TWICE NAT:

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Basic NAT does not work well for communication initiated from the Internet. Twice NAT allows a site to run servers. It requires the DNS to interact with the NAT device. Twice NAT fails if an application uses the IP addresses instead of Domain Name.

CAT:

Cable TV providers offering Internet services through Cable Modems propose it. It includes NAPT plus additional functionality. The additional functionality allows the cable operator to communicate with the CAT device, inspect values and control network access.

NAT AT HOME:

NAT is useful at a residence with Cable Modem or DSL connectivity as it allows the customer to have multiple computers at home without requiring an IP address for each of them. Instead a single IP address is used for all the computers. NAT software allows a PC to connect with the Internet and act as a NAT device at the same time.

Lecture No. 39 IP ROUTING (Part-1)

TERMINOLOGY:

The forwarding and Routing are two different concepts and explained as follows:

- FORWARDING
- ROUTING

FORWARDING:
It refers to datagram transfer. It is performed by host or router. It uses routing table.

ROUTING:

It refers to propagation of routing information. It is performed by routers. It inserts or changes values in routing table.

ROUTING:

It refers to propagation of routing information. It is performed by routers. It inserts or changes values in routing table.

TWO FORMS OF INTERNET ROUTING:

- STATIC ROUTING:
  It is one of the forms of Internet routing. In Static routing, the table is initialized when system boots and there is no further changes.
- DYNAMIC ROUTING:
  In dynamic routing the table is initialized when system boots. It includes routing software which

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learns routes and updates table. In this way continuous changes are possible due to routing software.

- STATIC ROUTING:
  It is used by most Internet hosts. The typical routing table has two entries. For the local network it has direct delivery and for the communication to some other network it follows the nearest default route.
- DYNAMIC ROUTING:
  It is used by IP routers. It requires special software which continuously updates the routing information. Each router communicates with neighbors. It passes routing information and uses Route Propagation Protocol to exchange the information with other routers.

Lecture No. 40 IP ROUTING (Part-2)

Routing in the Global Internet

The global routing table is a database maintained by IP on the switch route processor (SRP) module. It contains at most one route per protocol to each prefix in the table. Each of these routes is the best route known by a given protocol to get to that prefix.

AUTONOMOUS SYSTEM CONCEPT:

An autonomous system (AS) is a network or a collection of networks that are all managed and supervised by a single entity or organization. An AS is a heterogeneous network typically governed by a large enterprise. An AS has many different sub networks with combined routing logic and common routing policies.

CLASSIFICATION OF INTERNET ROUTING PROTOCOLS:

There are two broad classes of Internet Routing Protocol:

1. INTERIOR GATEWAY PROTOCOLS (IGPs)
2. EXTERIOR GATEWAY PROTOCOLS (EGPs)

INTERIOR GATEWAY PROTOCOLS (IGPs):

It is used among routers within autonomous system. The destinations lie within IGP.

EXTERIOR GATEWAY PROTOCOLS (EGPs):

It is used among autonomous systems. The destinations lie throughout Internet.

Lecture No. 41 IP ROUTING (Part-3)

ROUTES AND DATA TRAFFIC: Long

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Each ISP is an autonomous system that uses an Exterior Gateway Protocol to advertise its customer's networks to other ISPs. After an ISP advertises destination D, datagram destined for D can begin to arrive.

INTERNET ROUTING PROTOCOLS: Long

Following are the Internet Routing Protocols.

- "Border Gateway Protocol (BGP)
- "Routing Information Protocol (RIP)
- "Open Shortest Path First Protocol (OSPF).

BORDER GATEWAY PROTOCOL: Long

It is most popular Exterior Gateway Protocol in Internet. It has following

Characteristics:

1. "It provides routing among autonomous systems (EGP).
2. "It provides policies to control routes advertised.
3. "It uses reliable transport (TCP).
4. "It gives path of autonomous systems for each destination.
5. "Currently the EGP is of choice in the Internet.
6. "The current version is four (BGP-4).
7. "It provides facilities for Transit Routing.

ROUTING INFORMATION PROTOCOL (RIP): Long

It has the following characteristics:

1. "It is used for routing within an autonomous system (IGP).
2. "Hop Count Metric: RIP measures distance in network hops, where each network between the source and destination counts as a single hop.
3. "It uses UDP for all message transmissions.
4. "RIP is used over LAN. Version 1 of RIP uses hardware broadcast and version 2 allows delivery via multicast.
5. "It can be used to advertise default route propagation. An organization can use RIP to install a default route in each router.
6. "It uses distance vector algorithm.
7. "RIP allows hosts to listen passively and update its routing table.

Lecture No. 42 IP ROUTING (Part-4)

OSPF GRAPH:

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Networks and Routers can be illustrated using OSPF graph. Routers correspond to nodes in OSPF graph. Networks correspond to edges. The adjacent pair of routers periodically test connectivity and broadcast link-status information to area. Each router uses link-status messages to compute shortest paths.

OSPF AREAS:

OSPF allows subdivision of Autonomous System into areas. The link-status information is propagated within an area. The routes are summarized before being propagated to another area. It reduces overhead (less broadcast traffic). Because it allows a manager to partition the routers and networks in an autonomous system into multiple areas, OSPF can scale to handle a larger number of routers than other IGPs.

Lecture No. 43 IP ROUTING (Part-5)

INTERNET MULTICAST ROUTING:

Internet multicast routing is difficult because internet multicast allows arbitrary computer to join multicast group at any time. It allows arbitrary member to leave multicast group at any time. It also allows arbitrary computer to send message to a group.

IP MULTICAST SEMANTICS:

IP multicast group is anonymous in two ways:

1. Neither a sender nor a receiver knows the identity or the number of group members.
2. Routers and hosts do not know which applications will send a datagram to a group.

IGMP:

A standard protocol exists that allows a host to inform a nearby router whenever the host needs to join or leave a particular multicast group known as Internet Group Multicast Protocol (IGMP). The computer uses IGMP to inform the local router about the last application when it leaves.

FORWARDING AND DISCOVERY TECHNIQUES:

Routers not hosts have responsibility for the propagation of multicast routing information. The size and topology of groups may vary e.g. Teleconferencing often creates small groups and on the other side web casting can create a large group.

APPROACHES FOR DATAGRAM FORWARDING: LONG

In practice multicast protocols have followed three different approaches for datagram forwarding:

1. FLOOD-AND-PRUNE
2. CONFIGURATION-AND-TUNNELING

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3. CORE-BASED DISCOVERY

FLOOD-AND-PRUNE:

Flood-and-prune is ideal in a situation where the group is small and all members are attached to contiguous Local Area Networks. To avoid routing loops, flood-and-prune protocols use a technique known as Reverse Path Broadcasting (RPB) that breaks cycles.

CONFIGURATION-AND-TUNNELING:

Configuration-and-tunneling is ideal in a situation where the group is geographically dispersed. When a multicast datagram arrives, the routers at a site transmit the datagram on all directly attached LANs via hardware multicast.

Lecture No. 44 IP ROUTING (Part-6)

MULTICAST PROTOCOLS: LONG

Several multicast protocols exist. Some of the proposed protocols are:

- DISTANCE VECTOR MULTICAST ROUTING PROTOCOL (DVMRP)
- CORE BASED TREES (CBT)
- PROTOCOL INDEPENDENT MULTICAST_ SPARSE MODE (PIM-SM)
- PROTOCOL INDEPENDENT MULTICAST _ DENSE MODE (PIM-DM)
- MULTICAST EXTENSIONS TO THE OPEN SHORTEST PATH FIRST PROTOCOL (MOSPF)
- CLIENT-SERVER INTERACTION

DISTANCE VECTOR MULTICAST ROUTING PROTOCOL (DVMRP):

This protocol is used by the Unix program mrouted and the Internet Multicast backBONE (MBONE). DVMRP performs local multicast and uses IP-in-IP encapsulation to send multicast datagrams from one site on the Internet to another.

CORE BASED TREES (CBT):

A multicast routing scheme in which the protocol software builds a delivery tree from a central point. When a user joins a group, routers send a message toward the central point (i.e., the core) to search for the nearest participating router.

PROTOCOL INDEPENDENT MULTICAST_ SPARSE MODE (PIM-SM):

This is a protocol that uses the same approach as CBT to form a multicast routing tree. The designers chose the term protocol independent to emphasize that although unicast datagrams are used to contact remote destinations when establishing multicast forwarding. PIM-SM does not depend on any particular unicast routing protocol.

PROTOCOL INDEPENDENT MULTICAST _ DENSE MODE (PIM-DM):

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A protocol designed for use within an organization. Routers that use PIM-DM broadcast multicast packets to all locations within the organization. Each router that has no member of a particular group sends back a message to prune the multicast routing tree. The scheme works well for short-lived multicast sessions because it does not require setup before transmission begins.

MULTICAST EXTENSIONS TO THE OPEN SHORTEST PATH FIRST PROTOCOL (MOSPF):

A protocol designed for use within an organization. MOSPF builds on OSPF and reuses many of the same basic concepts and facilities.

CLIENT-SERVER PARADIGM:

It is used by all network applications. The passive program is called a server and the active program is called a client.

CHARACTERISTICS OF A CLIENT: LONG

The characteristics of a client are explained below:

- "Client is an arbitrary application program.
- "It becomes client temporarily.
- "It can also perform other computations.
- "It is invoked directly by the user.
- "It runs locally on the user's computer.
- "It actively initiates contact with a server.
- "It contacts one server at a time.

CHARACTERISTICS OF A SERVER:

The characteristics of a server are explained below:

- "It is a special-purpose, privileged program.
- "It is dedicated to provide one service.
- "It can handle multiple remote clients simultaneously.
- "It invoked automatically when system boots.
- "It executes forever.
- "It needs powerful computer and operating system.
- "It waits for client contact.
- "It accepts requests from arbitrary clients.

Lecture No. 45 COURSE REVISION

COURSE SUMMARY

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